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Children have fundamental right to access Internet: SC

Krishnadas Rajagopal
NEW DELHI

The Supreme Court has held that children, like all individuals, have a fundamental right to access and use the Internet under Articles 21 and 19(1)(a) of the Constitution.

A Bench of Justices J.B. Pardiwala and Vinod K. Chandran, in its order, cautioned that blanket bans on children's online access were "unproven measures of proven disproportionality". It said the safety of children and their freedom were not at odds with each other and that an atmosphere of hostility and stringent restrictions could stop children, enticed or threatened online, from reporting such incidents to parents or guardians for fear of reprisal for using the Internet.

FULL REPORT ON
» PAGE 9

A ND-NDE

Law is shaped by voices that do not always harmonise: CJI

Law will give you 'joy of service and honour of upholding the Constitution', says CJI Kant at NLSIU convocation, which had been postponed over students' support for their NALSAR counterparts

The Hindu Bureau
BENGALURU

Chief Justice of India (CJI) Surya Kant on Saturday said that the purpose of legal education "rests in the invigorating experience of encountering those who see the world differently than we do".

Chief Justice Kant, also the Chancellor of the National Law School of India University (NLSIU), Bengaluru, was speaking at the 34th convocation of the university.

The convocation was earlier scheduled to be held on September 12. However, it was postponed after students and alumni of the NLSIU came out in support of students at NALSAR, Hyderabad, who had opposed the Chief Justice's presence at the convocation over his remarks about the country's youth.


CJI Surya Kant with D.K. Shivakumar, actor Arundhati Nag and other dignitaries at the NLSIU in Bengaluru. ALLEN EGENUSE J.

However, the convocation went off peacefully on Saturday, with no protests or untoward incidents.

Elaborating on the idea of law as encountering different points of view, the CJI said, "Within the halls, we sit beside people whose backgrounds, beliefs, and interpretations challenge our own. Each time we listen to perspectives that diverge from ours, we are reminded that the law itself is a living conversation, shaped by voices that do not always harmonise, but together create the symphony of justice. It is your relentless pursuit of honesty which will cement your professional reputation."

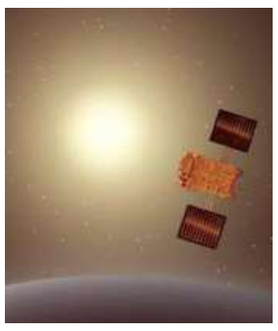
Asking the students to uphold the Constitution and its values, he said, "Law will change your patience, test your ethics, and call upon the deepest reserves of your courage, but it will also reward you with a singular joy of service, the satisfaction of righting a wrong, the pride of defending liberty, and the honour of upholding the Constitution."

Bengaluru scientists study solar eruptions with data from Aditya-L1, Proba-3 satellites

The Hindu Bureau
BENGALURU

In a first, scientists from Bengaluru and their international collaborators have analysed scientific observations of solar eruptions and studied the coronal mass ejections (CMEs) by using data derived from India's Aditya-L1 mission and Europe's Proba-3.

Scientists at the Indian Institute of Astrophysics (IIA) combined observations from the Visible Emission Line Coronagraph (VELC) payload onboard the Aditya-L1, and the Association of Spacecraft for Polarimetric and Imaging Investigation of the Corona of the Sun (ASPIICS) payload onboard the Proba-3 to study the mass, density, and energetics of these CMEs in the region very close to the Sun.


A file photo of Aditya-L1 entering the Halo Orbit. ISRO

The coronagraph onboard Aditya-L1 (VELC) and Proba-3 (ASPIICS) conducted joint observations of almost the same part of the solar corona in two different types of observations.

The Indian Space Research Organisation (ISRO) stated that the joint observations provide deeper insights into the CME evolution than observations from a single spacecraft.

"The ASPIICS observations provide sharp images that enable accurate estimates of a CME's shape, dynamics, and mass. The VELC spectroscopic observations provide information on the CME gas at about two million degrees. This gives independent estimates of the CME's density and thermal energy. The two instruments share a common field of view in the low corona. The team found that only two CMEs, on September 14 and 16, were recorded simultaneously by the VELC and the ASPIICS from July to December 2025. In both cases, the VELC caught the eruptions at an early stage, demonstrating how the two instruments complement each other," the ISRO stated.

The Aditya-L1, which is India's first dedicated solar mission, was launched on September 2, 2023. The Proba-3, a mission of the European Space Agency (ESA), was launched from Sriharikota on December 5, 2024.

The two spacecraft are about 150 metres apart, and one casts a shadow on the other.

Children have fundamental right to access Internet: SC

Court says blanket bans are unproven measures of proven disproportionality; favours parental oversight, age-appropriate safeguards, dedicated 'child mode' and better efforts from stakeholders

Krishnadas Rajagopal
NEW DELHI

The Supreme Court has held that children, like any other individual, have a fundamental right to access and use the Internet.

"We have no hesitation in saying that the right to access and use the Internet is a fundamental right of every individual under Article 21 read with Article 19 (1)(a) of the Constitution. Our children, irrespective of their age, enjoy the same rights, and to deny the same altogether would be absurd," a Bench of Justices J.B. Pardiwala and Vinod K. Chandran held in an order.

The court said denying children this right would be akin to saying that individuals below a certain age did not have a conscience worth protecting or freedom of speech worth expressing.

The safety of children and their freedom were not at odds with each other, it said. "Our children are entitled to both," the Bench observed.



The court said keeping children away from social media would not reduce incidents of online abuse reported in the media. FILE PHOTO

Issuing a slew of directions for protective mechanisms, restrictions drawn from other jurisdictions and grievance outlets, the court emphasised that intelligent restraints, rather than blanket bans, were the need of the day.

The court took a practical view that children did not cease to use the Internet merely because the law prohibited them from doing so. "They continue to use the same, albeit without the knowledge of their parents or the authorities. They move to other platforms and websites that are unregulated and even more unsafe," it said.

An atmosphere of hostility and stringent restrictions could stop children, enticed or threatened online, from reporting such incidents to parents, guardians and trusted adults for fear of reprisal for using the Internet. "This would significantly undermine the safety of our children, who are already apprehensive about confiding such incidents, even without such restrictions, and defeat the object of the POCSO Act," the court said.

Even otherwise, it said keeping children away from social media platforms would, by no stretch of imagination, reduce incidents of online abuse reported in the media.

"All that we are trying to say is that any blanket restriction on access to the Internet by children is nothing more than unproven measures of proven disproportionality. No jurisdiction has so far been able to demonstrate that children are any safer on account of such measures. Its efficacy is at best contested, and without question, not a substitute for the initiative and efforts required from all the stakeholders, be it the intermediaries, the law enforcement agencies, the Ministry of Electronics and Information Technology or the courts," the Bench observed.

The Bench, however, said that their observations were not against age-based restrictions on online access. "We acknowledge that some form of safeguards are necessary for the safety of our children online and the moderation of the content accessible at their fingertips," it said.

The court suggested a system that placed greater emphasis on parental consent, guidance and supervision of children's access to social media platforms, along with a dedicated "child mode" for such accounts, with measures proportionate to the objective of ensuring their safety.

CRPF may bring back Naik ranks to boost morale

Press Trust of India
NEW DELHI

The Lance Naik and Naik ranks were abolished in the force in 1997

Central Reserve Police Force (CRPF), the country's largest paramilitary force, is mulling reintroducing the military ranks of Lance Naik and Naik, 30 years after they were abolished, to address delayed promotions and accommodate Agniveers retiring from the Army, officials said on Saturday.

The CRPF headquarters has asked hundreds of field formations of the force across the country to conduct stakeholder meetings or 'Sainik Sammelans' and obtain the views of the personnel in this context in two days, they told *PTI*.

The Lance Naik and Naik ranks were abolished in the force in 1997 and merged with the police ranks of constable and head constable through a Ministry of Home Affairs (MHA) order.

The two ranks are still operational in the Army.

According to the hierarchy, a jawan who joins the force as a constable gains the first promotion of Lance Naik in five years, followed by the promotion to Naik in 10 years and subsequently head constable in 20 years.

Currently, the about 3.25 lakh personnel-strong CRPF is facing acute stagnation in promotions in these ranks, as a constable gets his first promotion in 22-25 years on average.

This stagnation in one rank has led to grievances among the jawans.

The officials said the aim of reintroducing the Lance Naik and Naik ranks is to provide two more promotions to the constabulary before they reach the head constable rank, which will not only help address their grievances but also boost their morale.

The move is also aimed at making the CRPF service attractive for the Agniveers from the Army, whose first batch will begin retiring later this year, and will be eligible to join the Central Armed Police Forces (CAPFs), including the CRPF, the officials said.

Agniveers are short-term military personnel recruited into the three forces for four years under the Centre's Agnipath scheme.

FSSAI proposes pesticide residue restrictions for tea

The Hindu Bureau
NEW DELHI

The Food Safety and Standards Authority of India (FSSAI) has proposed maximum residue limits (MRLs) for two pesticides, acetamiprid and imidacloprid, in tea, said a release issued by the Central Government on Saturday.

The draft Food Safety and Standards (Contaminants, Toxins and Residues) Amendment Regulations, 2026, published in the Gazette of India on Friday, propose MRLs of 20 mg/kg (20 parts per million) for acetamiprid and 50 mg/kg (50 ppm) for imidacloprid in tea.

The proposed limits are intended to establish regulatory benchmarks for pesticide residues in tea and facilitate compliance, monitoring and enforcement of food safety requirements, subject to the finalisation of the draft regulations. Notably, the proposed acetamiprid limit is lower than the corresponding limits of 30 ppm in Japan and 50 ppm in the United States. The proposed imidacloprid limit is also based on international Codex standards.

It added that India's average pesticide use in agriculture, at less than 0.5 kg per hectare, is significantly below the estimated global average of 2.38 kg per hectare. "It is also lower than the levels reported in several countries, including Colombia, Brazil and the United States," the release said.

The limits are intended to provide greater regulatory clarity to tea growers, processors, and businesses, it said.

FSSAI moves to ban added sugar in infant food items

Bindu Shajan Perappadan
NEW DELHI

The Food Safety and Standards Authority of India has proposed prohibiting the addition of sucrose and fructose in food products intended for infants up to six months of age.

The draft Food Safety and Standards (Foods for Infant Nutrition) Amendment Regulations, 2026, published in the Gazette of India on Friday proposes to amend the existing regulations governing the nutritional composition, quality and safety of infant nutrition products.

"The first six months are a critical period for infant growth and development, making appropriate nutrition particularly important. The proposed restriction aims to ensure that products intended for this age group meet their nutritional requirements without the addition of sucrose and fructose," the government maintained in a release issued on Saturday.



The government release mentions early exposure to added sugars as a public health concern. ISTOCK PHOTO

It further mentions that early exposure to added sugars is a public health concern because of its potential implications for the development of dietary preferences and long-term dietary habits. "The proposal also complements public health recommendations advocating exclusive breastfeeding during the first six months of life," the release said.

Under the existing Food Safety and Standards (Foods for Infant Nutrition) Regulations, 2020, lactose and glucose polymers are the preferred sources of carbohydrates in infant nutrition products. Sucrose and/or fructose may be added when required as a carbohydrate source, subject to prescribed limits.

India adapts fast as West Asia simmers

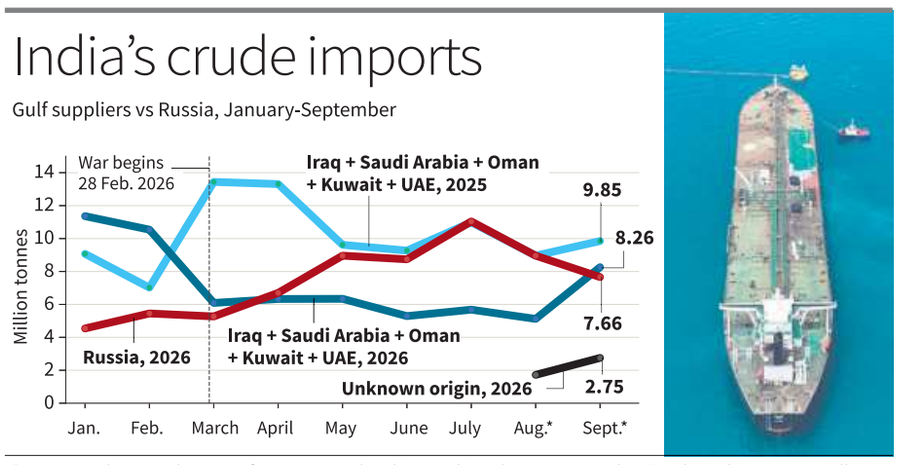
Kpler figures indicate in September Iraqi crude exports to India tripled over August; shipping industry sources said India was buying large parcels from many nations depending on availability; the same strategy would be deployed if Gulf crude disruptions occurred in October also

M. Kalyanaraman
CHENNAI

In September, the Consultative Shipping Group, a group of leading maritime nations, said conflicts such as over the Strait of Hormuz and Black Sea are "not episodic shocks; they are signals of a structural shift in the operating environment of global trade."

Shipping routes are increasingly instruments of leverage and risk."

September saw shipping attacks intensify across the Black Sea even as attacks on ships across the Strait of Hormuz continued but with less intensity. The UN general assembly saw major world figures gathering in New York. Iranian Foreign Minister Abbas Araghchi spent almost a week in New York, holding talks with the U.S. government through intermediaries. The Iranian President addressed the general assembly and spoke to an American television channel. But there was no sign of an immediate truce.



**August and September 2026 figures are Kpler ship-tracking data, converted at 7.33 barrels per tonne. All others data, including the 2025 figures for August and September: Ministry of Commerce and Industry trade statistics (TradeStat), HS27090010. Unknown origin is a Kpler category; it has no TradeStat equivalent*

September was the month Gulf oil exports rebounded. Since the war started, Saudi Arabia had been sending nearly half of its typical crude oil exports via its western ports on the Red Sea by piping the oil across the country.

Despite the attack on that pipeline disrupting the flow in September, Saudi Arabia was able to quickly ship out from Ras Tanura on the Persian Gulf and eventually through ship-to-ship transfers off Oman. Overall, oil exports from Gulf countries plummeted when the war started and crawled up to some 60% in June-July, rebounded to 80%-100% of pre-war levels in September.

Pivot to Gulf

As Russian imports dipped, India was able to pivot toward Gulf oil in September. Kpler figures, which are typically off eventual Ministry of Commerce figures by a few percentage points, indicated that in September Saudi crude export to India rose 50-60% over August.

Iraqi imports almost tripled while Kuwaiti and Omani imports to India ramped up, too. Gulf oil imports crossed Russian shipments to India for the first time since March.

"Gulf crude oil imports to India that was at 1.2 million barrels a day rose to 2.4 million barrels per day in September. Flows across the Strait of Hormuz, Red Sea via Bab el-Mandeb, Red Sea via Suez Canal, as well as the land bypass of Strait of Hormuz to Fujairah helped almost catch up with pre-war levels," says Pankaj Srivastava, senior VP, Rystad Energy.

End of September saw ship attacks ramping up, turning the most intense since the war started. More than a dozen attacks have been reported already in October and two ships sunk. Mr. Srivastava is, however, confident that Gulf crude exports would be comparable to September via multiple channels.

Shipping industry sources said India was buying large cargoes from many nations as per availability. The same strategy would be followed if Gulf oil disruptions occurred in October. Government figures show occasional purchases of over 1 million tonne/month from countries such as Brazil, Angola, Nigeria, Venezuela and the U.S.

Indian IT exports: Deliver code, chase cash

Balasubramanyam C.P
CHENNAI

A Bengaluru technology company says it is still waiting for $6,72,102 from an overseas customer for work completed more than 19 months ago, highlighting the financial risks smaller Indian technology exporters face when international clients fail to pay.

Thiyagu Ganesan, founder of Ticvic Technologies Pvt Ltd., sought the Karnataka government's intervention. His petition says the company completed the contracted work and the customer's technical team tested and confirmed that it was functioning.



Not easy: Cross-border recovery can become prohibitively expensive for smaller companies.

The contract shared by Ticvic shows a second-phase project worth $6,72,102, with payments linked to milestones and customer acceptance. In an email dated May 29, 2024, a customer representative wrote: "I tested it and found that the NAT function is fine."

Mr. Ganesan said Ticvic had received payment for an earlier project and, trusting the relationship, proceeded with the second phase without the expected upfront payment. He personally mortgaged land and pledged gold to keep all 47 staff paid, expecting to redeem the assets once the customer settled the dues. The company's operations have since shrunk significantly, leaving only a few employees.

It says it approached China's Ministry of Commerce, CCPIT Mediation Center and Indian Embassy in Beijing, but the dispute remains unresolved.

Recourse for exporters

Phil Fersht, CEO and chief analyst of HFS Research, said the remedies depend largely on the contract. Companies can pursue debt recovery, arbitration or litigation, enforce letters of credit or bank guarantees, or claim export-credit insurance where cover was arranged beforehand.

However, cross-border recovery can become prohibitively expensive for smaller firms. "Pursuing a $5,00,000 or $1 million claim via another country's legal system can take considerable time and money," he said. Once software is delivered and accepted, "the supplier has already surrendered much of commercial leverage."

The Directorate General of Foreign Trade's trade-dispute mechanism covers non-payment involving goods, services and technology while the Export Credit Guarantee Corporation offers insurance for software and technology exports. But prevention remains crucial.

IISER researchers make solid material that releases two fuel gases

Vasudevan Mukunth

Scientists at the Indian Institute of Science Education and Research, Thiruvananthapuram, have developed a chemical compound that releases hydrogen and methane when heated moderately. Per the team, the work is a step towards new, better methods to store and transport these fuel gases.

The findings were published in *Science Advances* on October 7.

Both in India and around the world, hydrogen is emerging as an important new fuel because its combustion does not directly release carbon dioxide. However, because of its very low density, hydrogen is very difficult to store and transport and demands the use of expensive technologies.

Scientists have previously proposed blending hydrogen with methane as an intermediary step: while hydrogen burns faster than methane, methane can reduce the risk of flames travelling back into burners or triggering violent detonations. They have also devised ways to store the two fuels as compressed gases and solid materials that can release one or the other gas.

In the new study, researchers developed a compound of magnesium, zinc, hydrogen, and carbon called magnesium hydridomethylzincate. When the compound was heated in solution to 100° C, it released hydrogen – and when its crystals were heated in the absence of a solvent, at 80-110° C, it released both hydrogen and methane. "The broader significance of the work is that organometallic compounds could potentially provide molecular platforms for the storage and controlled release of fuel gases," chemistry professor and study lead Ajay Venugopal told *The Hindu*.

The team established that methane emerged before hydrogen, indicating the compound was releasing the gases through successive chemical reactions. Further analysis also revealed that each unit of the compound's paired molecular structure could produce two molecules each of hydrogen and methane.

Since magnesium hydridomethylzincate is a solid at room temperature, it could offer an alternative to transporting methane and hydrogen as compressed gases. This together with the relatively low temperature required to liberate the gases merit more research on the compound, the paper said.

More work is needed since, as the team has said, the compound is sensitive to moisture and it is not clear if it can be 'replenished' once the gases are 'depleted'.

Although we are yet to demonstrate reversibility in our system, the study provides a starting point for exploring molecular compounds that can reversibly uptake and release fuel gases using main-group elements," Prof. Venugopal said.

As inscrutable AI proofs emerge, mathematicians wait in the dark

Powered by arrays of graphics processing units, large language models code better than experienced coders and write award-worthy novels; but mathematicians are becoming more concerned about what it bodes for their future

Jacob Koshy
NEW DELHI

The French mathematician Hugo Duminil-Copin, who won the Fields Medal in 2022, said in a video call of the French Mathematical Society on October 8 that years of his research agenda had been "wiped out" thanks to a tumult of mathematical proofs and results released by OpenAI a day prior.

Sixteen of these results answered questions Duminil-Copin had spent years trying to solve. He said his field has been "annihilated".

OpenAI released some 700 manuscripts from an unreleased model, many with potentially field-changing implications for mathematical research. Last month, OpenAI had claimed a proof, certified by the Lean proof-checking software, that the Navier-Stokes equations of fluid mechanics are incomplete, a problem of similar potential.

But while much of it has been machine-checked, been refereed or even read by humans. Mathematicians build skyscrapers of abstraction but they do not become 'true' until every logical nook and cranny has been thoroughly vetted.

Gifted practitioners with the dazzling ability to see connections between phenomena as different as cowlicks and coconuts must still wade through the quicksand of proof.

Terence Tao of the University of California, Los Angeles, who co-signed an influential statement with other Fields Medalists called 'A Severe Misalignment of AI in Mathematics', has argued that the journey to proof is nearly as important as the destination. If an advanced AI model produces a drug cocktail that a clinical trial confirms can cure cancer, he has asked, wouldn't you want at least one human expert to understand the mechanism before it went into your bloodstream? Verification, according to Tao, is necessary but insufficient because somebody must still understand what has been verified.

Chandrashekhar Khare, also of UCLA, told *The Hindu* by email that AI's ability to make connections between seemingly unrelated ideas did appear to be one of its "dazzling abilities".

"I do not know in what sense an LLM [large-language model] understands a mathematical argument... If it correctly establishes its conclusion from its assumptions, it is a proof regardless of how it was produced. Fluent mathematical prose alone, however, does not establish correctness," he explains.

"In some ways, I find it reassuring that they seem to be drawing extensively on ideas and methods developed by human mathematicians, combining them in extraordinarily creative ways and pushing them further than we had managed. Their progress seems to grow out of the mathematics people have created, rather than bypassing it through some wholly unfamiliar method."

"Only time will tell how far the new AI-generated solutions have ... continuing mathematical influence," he adds.

That said, the opacity of AI-generated proofs also bothers Dr. Khare.

"Mathematicians generally discuss ideas freely and make their work available to others. There is competition, and occasionally priority disputes, but openness is an important ideal of the subject. Tremendously powerful proprietary models, whose availability is controlled by companies and whose use depends on financial resources, introduce an element that sits uneasily with this ideal."

At the Indian Institute of Science, Bengaluru, Mahesh Kakde, recently awarded the Ramanujan Prize for work on noncommutative Iwasawa theory and the Gross-Stark and Brumer-Stark conjectures, which have implications for one of number theory's oldest ambitions, called Hilbert's 12th problem, says: "My AI concerns are much wider than mathematics. Firstly, I have concerns about the great environmental costs of AI. Secondly, I am concerned about the concentration of power in the hands of a few. In the last 50 years, every advancement in digital technology was predicted to democratise various things ... but by and large, it has concentrated power in the hands of a few. AI will most likely exacerbate all these problems."

Lean, he adds, is effective "but with caveats". The kernel that applies the rules may have a bug or a definition or theorem can be mistranslated into Lean. "Since LLMs seem to want to accomplish a task at any cost, even by finding a kernel bug to get a certificate, I wouldn't put even deliberate mistranslation beyond them."

The training of future mathematicians worries both.

'Will be left far behind'

"Training PhD students usually involves giving a problem that is well within reach. ... This is how a student learns to survey the literature, get familiar with the mathematical objects, and develop their own intuition about them. If these can just be put in LLMs to get a solution, then they lose out on building these important skills," Dr. Kakde says. "Secondly, if LLMs keep solving hard open problems regularly, then we will be left far behind."

"A PhD should," Dr. Khare adds, "help a student become an independent mathematician with a point of view of their own."

Maths and mysteries

'Only time will tell how far the new AI-generated solutions have continuing mathematical influence'

- OpenAI recently shocked mathematicians by releasing numerous automated proofs and claiming to have solved historic problems
- While many complex proofs have been verified by machines, they still want for actual human understanding
- These machines have combined ideas creatively but whether they operate with true awareness or logical understanding is unclear



Chandrashekhar Khare

- Experts worry that highly opaque proprietary intelligence models fundamentally conflict with the ideals of open mathematics
- They are concerned that relying on automated solutions could hamper students' ability to acquire essential skills, including developing intuitions
- They are also thinking about the environmental costs; one suggests corporations may simply be benchmarking their systems for impending IPOs



Hugo Duminil-Copin

CREDIT: INFOSYS PRIZE FOUNDATION, IHES



Free database 'shows' how drivers of ageing team up

Understanding how and why our bodies age has often meant sifting through scattered data. In response, researchers from institutes in Chennai and Mumbai have developed a free database called AgingHallmarksDB. The platform brings together 3,111 genes associated with 11 major core biological processes driving how we age. By consolidating this data, the team behind the database hopes to make it easy for scientists and the public alike to visualise how these complex processes connect.

The tree bridges that help gibbons 'cross' highways



SPEAKING OF SCIENCE
D. Balasubramanian

The Hoolock gibbon is India's only ape other than us humans. Found in Northeast India, mostly to the south of the Brahmaputra river, the western and eastern hoolock gibbon populations have declined by 90% in the last few decades. The main reason for this has been the fragmentation of their habitats – the range that individuals can freely roam over – by roads and railway lines.

The gibbons are particularly vulnerable to such fragmentation because of their unusual body structures. Adults are typically three feet tall but have five-foot long arms. They are very vulnerable when walking on land: their awkward gait makes them easy targets for wild dogs and leopards. The gibbons have adapted to an arboreal lifestyle, using their long arms and the natural spring action of tree branches to move high above the ground with a rapid grace.

Yet this also requires a continuous canopy of trees. A road or a rail line completely cuts off one section of a forest for them. These gibbons mate for life and mated pairs ferociously guard their territory. Linear infrastructure effectively creates isolated pockets of gibbon populations. Smaller territories support fewer gibbons, and over time this leads to genetic isolation, inbreeding, and a decline in numbers.

The problem is not faced by gibbons alone.



A male western Hoolock gibbon hangs from the lower branches of a tree in the Hoollangapar Gibbon Sanctuary. KRUNAL DESAI (CC BY-SA)

The lion-tailed macaque's habitat around the town of Valparai in Tamil Nadu is fragmented by coffee and tea estates, and interconnecting roads lying between stretches of rainforest.

Around the world too, the slow loris in Indonesia and the three-toed sloth in Central America have faced fragmentation catastrophes. Both these animals are notoriously slow movers, preferring to move while hanging upside down on branches.

Hoolock gibbon populations began to be fragmented when railway networks were expanded into Eastern Assam in 1887. One such fragment eventually became the Hoollangapar Gibbon Sanctuary, which has a woefully small population of about 130 western hoolock gibbons, out of a total of only 3,000-5,000 in Northeast India. Conservation efforts began in earnest long after the railway lines were operational, led by local organisations such as Aranyak.

To create a natural bridge over the railway line, a careful and dense planting of trees on either side of the track was undertaken. About 15 years later, the canopies of trees on either side of the track came together to form a bridge that the apes could swing over. And even before the canopy bridge was established, gantry bridges were put up.

Efforts to help arboreal creatures find their kin and supplies of food have been successful at other locations, too. The giant Indian squirrel is more frequently seen as roadkill than in the wild. This magnificent creature, which has a body that is over a foot in length and an even longer tail, prefers to live in the canopies above 30 feet, and has been aided by braided rope bridges installed at many places, from the Melghat Tiger Reserve in the Satpuras in Maharashtra to the Srivilliputhur Grizzled Squirrel Sanctuary in Tamil Nadu.

Other land animals too need measures to prevent habitat fragmentation. Heavily populated countries such as the Netherlands are at the forefront of this. This country has an animal overpass that is nearly a kilometre long and spans a highway, a rail yard, and a golf course. Our increasingly busy highways surely need such facilitators for our wildlife.

(The article was written in collaboration with Sushil Chandani, who works in molecular modelling)
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Question Corner

Eye to eye

Why are the whites of human eyes white?

The tough outer covering of the eyeball is called the sclera. In humans, the sclerae (in both eyes) are white. This allows someone standing a distance away to easily say which way a person is looking.

Scientists are not sure why human sclerae are white. Chimpanzees may have white, brown or darker sclerae. Bonobos and Sumatran orangutans have light brown sclerae. On the other hand, researchers have documented pale sclerae in golden langurs and southern pig-tailed macaques and some leopards and sea lions.

One leading idea is called the cooperative eye hypothesis, which says we evolved conspicuous eyes because our ancestors could benefit from signalling to others which way they were looking. However, a review published in 2025 in *Biological Reviews* said the hypothesis has not accumulated enough evidence. For example, in 2023, scientists found that the chimpanzees they were studying in Uganda were able to say where other individuals were looking even with darker sclerae.

Readers may send their questions / answers to science@thehindu.co.in

Why have Central banks around the world began hiking rates?

How is this cycle different from 2022? What is happening in U.S. Treasury yields? Why should India be concerned?

Gowri Lakshmi

The story so far:

Geopolitical tensions in West Asia, elevated crude oil prices, rising U.S. Treasury yields, and persistent inflation are complicating the global monetary policy outlook. After a period of monetary easing, central banks may have less room to cut interest rates further, with renewed inflationary pressures potentially forcing some to reconsider their policy stance.

What makes the current cycle different from 2022?

The key difference from the previous rate-hiking cycle is the starting point. In 2022, several major central banks began tightening from exceptionally low interest rates. The U.S. Federal Reserve and European Central Bank were emerging from prolonged periods of accommodative policy with benchmark rates at 0% or in the negative territory, while the Bank of Japan maintained a different policy trajectory. India's benchmark repo rate stood at 4% at the beginning of its tightening cycle.

Today, borrowing costs in many economies are already restrictive, while fresh inflationary pressures threaten to weaken economic growth. Sticky inflation makes the challenge more difficult, as policymakers must contain rising prices without unnecessarily slowing economic activity.

For India, higher crude prices increase the import bill, while a weaker rupee makes dollar-denominated oil imports more expensive. These pressures can push up domestic prices.

Starting from low rates gave many central banks greater scope to raise borrowing costs. However, higher rates still increased loan repayments and financing costs for households and businesses.

Today, central banks may have less room to tighten without weakening consumption, investment, and employment. At the same time, renewed energy-price shocks could prevent them from cutting rates as quickly as they otherwise would. This creates a difficult trade-off: easing too soon could prolong inflation, while keeping rates high for too long could undermine growth.

What is the global economic outlook?

International Monetary Fund Managing Director Kristalina Georgieva has said that the global economy has remained resilient despite supply-chain disruptions and higher oil prices. However, the prolonged conflict in West Asia threatens to test that resilience, particularly if energy prices remain elevated. Tariff uncertainty from Washington, rising public debt, and the AI investment boom are adding to economic uncertainty. According to the World Bank, global economic growth is projected to slow to 2.6% in 2026 from 2.9% in 2025.

What are the main constraints on economies?

Higher crude oil prices remain a major concern, particularly for energy-importing economies such as India. Oil is a critical input for transport, manufacturing, and other industries. Rising energy prices increase production and logistics costs, some of which businesses pass on to consumers. For India, the impact extends beyond domestic inflation. Higher crude prices increase the import bill, while a weaker rupee makes dollar-denominated oil imports more expensive. These pressures can worsen the current-account balance and push up prices.

What is happening in U.S. Treasury yields?

U.S. Treasury yields have risen since late February. Bond prices and yields move in opposite directions: when existing bond prices fall, their yields rise. Higher Treasury yields can increase borrowing costs for governments and businesses. Rising yields reflect several factors, including expectations of interest rates remaining higher for longer, inflation uncertainty, and the additional compensation investors demand to hold long-term government debt. Increased government borrowing can add to these pressures. The AI boom is also influencing capital markets. Technology companies are investing heavily in data centres and computing infrastructure, increasing their financing requirements and competing with other borrowers for capital.

Why should India be concerned about U.S. Treasury yields?

U.S. Treasuries serve as important benchmarks for global financial markets. Their yields influence borrowing costs and valuations across countries, including the pricing of corporate bonds, mortgages, and emerging-market debt.

For India, sustained pressure on the rupee could amplify the impact of elevated crude prices. A weaker currency increases the domestic cost of energy imports, potentially raising fuel, transport, and production costs. This could make it harder for the Reserve Bank of India to ease monetary policy.

How does the U.S.-Russia diesel deal impact India?

What does the U.S.'s diesel deal with Russia entail and what is President Trump's reasoning behind it? Why are some Republicans and Democrats unhappy with the deal? How does the deal impact India and its trade deal with the U.S.? Is there an 'energy ceasefire' between Russia and Ukraine?

T.C.A. Sharad Raghavan

The story so far:

Less than a month after signing into law an Act that would impose tariffs of up to 100% on countries importing large amounts of Russian oil and gas, U.S. President Donald Trump announced a deal with Russia on October 10, 2026, that would allow it to sell diesel to the U.S. and the global market. The next day, Mr. Trump also announced an 'energy ceasefire' between Russia and Ukraine. This has implications for India, its energy sector, and its trade.

What is the deal with Russia on diesel?

On October 10, Mr. Trump took to Truth Social to say he has concluded "a highly successful discussion" with Russian President Vladimir Putin in which they agreed that Russia would "immediately" supply over 3,00,000 tonnes of diesel to the U.S. and to the global market, another 5,00,000 tonnes in November, and 1 million tonnes "immediately thereafter".

Further, based on the condition of Russia's diesel refineries, he said the country would deliver another 3 million tonnes of diesel "within a short period of time" following this schedule.

Mr. Trump's reasoning for this move was that it would bring down fuel prices for the U.S. and the world. In fact, in response to criticism on the deal, he again took to Truth Social to say that he had "made the right choice for the American People – ALWAYS, AMERICA FIRST!".

The latest U.S. deal with Russia on diesel is expected to further complicate matters since the U.S. is indulging in the very actions it is looking to penalise India for.



U.S. President Donald Trump and Russian President Vladimir Putin during a press conference in Anchorage, Alaska, U.S., in August 2025. REUTERS

What are the complications with the deal?

On September 18, Mr. Trump signed into law the Lindsey O. Graham Sanctioning Russia and Iran Act of 2026. This Act received remarkable bipartisan support from Democrats and Republicans in both the Senate and the House of Representatives. It aims to clamp down on Russia's funding for its war in Ukraine through sanctions on its leadership and companies, and tariffs on its major fuel trade partners.

Now, leaders on both sides of the political spectrum are saying that the President's latest diesel agreement with Russia undermines the very foundation of the law. "Through the Lindsey O. Graham Sanctioning Russia and Iran Act, we gave the president significant authorities [sic] and leverage against China and Russia to bring Putin's war to an end with a negotiated settlement," House of Representatives member and Republican Michael McCaul of Texas said in an X post. "Unfortunately, while I understand the desire to bring down diesel prices, I am concerned the lifting of sanctions on Russian oil will only fund the Kremlin's war machine – emboldening more violence and destruction, as we have seen in recent days."

A further complication is that the Act threatens tariffs of up to 100% on the largest importers of Russian oil and gas. China and India are by far the biggest importers of Russian energy. However, Mr. Trump on Saturday announced that he has waived the tariffs associated with the Russia Sanctions Act. The waiver is not permanent, meaning these tariffs could be imposed in the future.

What are the implications for India?

India has been trying to conclude a trade deal with the U.S. since February 2025. However, each time progress is made, the U.S. makes a change to its tariff or trade policies that scuppers discussions. This time will be no different.

While talks were gathering steam in June last year over a 'mini deal' on tariffs, the escalation of U.S. tariffs on India to 50% in August 2025 stalled those talks. The reason for that escalation was India's imports of Russian oil.

Talks resumed in October 2025. By February 2026, the leaders of the two countries issued a joint statement saying they had agreed to a framework for an interim agreement on trade. The deadline for this deal was unofficially set as March-April 2026.

However, in March, the U.S. launched investigations into India and several other countries to determine whether they had done enough to prevent imports of goods made using forced labour and whether they were using structural excess capacity to harm the U.S. economy. While the forced labour investigation resulted in 10% tariffs on India in July, the investigation into excess capacity is yet to be completed. Once again, the deal talks slowed down, a situation exacerbated by the passage of the Russia Sanctions Act in September.

The latest U.S. deal with Russia on diesel is expected to further complicate matters since the U.S. is indulging in the very actions it is looking to penalise India for. While historical data show that India has largely toed the line on Western sanctions and prohibitive tariffs imposed on Russia, the situation this time could be different. Even if the U.S. rescinds its tariff waiver in the near future, fuel supply constraints arising from the conflict in West Asia could mean that India is unwilling to turn away from Russian oil.

What are the implications of the 'energy ceasefire'?

Mr. Trump did not reveal any details about the ceasefire, its timeline, duration, or extent. In fact, in response to his announcement, Ukrainian President Volodymyr Zelenskyy said that the ceasefire was "news for me", implying he was not aware of the discussions. However, he later said that he was willing to cooperate.

Ukraine has been attacking Russian refineries to the extent that Russia has been forced to import refined fuel from its partners despite itself having vast amounts of crude oil.

Russia imported a record 1,72,000 tonnes of oil products in August, according to a report by the Centre for Research on Energy and Clean Air. The report added that India accounted for 70% of these imports by Russia.

Should an energy ceasefire come into effect, and depending on its duration, Russia will likely be able to repair its refineries and begin its own supplies of fuel. This would mean it would no longer require fuel from India.

However, that ceasefire may remain just a statement. Days after the announcement, Mr. Zelenskyy took to social media to say that "last night, we did not attack any energy facilities on the territory of Russia... while Russia attacked our gas stations, energy infrastructure, and other facilities". He added: "The message from Russia was very clear – they don't know anything about the energy ceasefire."

Why was actor Imran Khan asked to leave India?

Who is eligible for an Overseas Citizen of India card? What privileges do OCI cardholders have in India and what activities and professions are they barred from? Do they have political rights? Can Imran Khan challenge the notice given to him? Can OCI registration be cancelled?

Rizmi Lia M.

Actor Imran Khan, a U.S. citizen and Overseas Citizen of India (OCI) cardholder, has been asked to leave India by October 28 after participating in a protest against Chief Election Commissioner Gyanesh Kumar in Mumbai. The Foreigners Regional Registration Office (FRRO) issued the notice following a complaint by Bharatiya Janata Party (BJP) leader Kirit Somaiya, who alleged that Mr. Khan had violated the law by participating in the demonstration. The notice invokes Sections 7(2)(c) and 29 of the Immigration and Foreigners Act, 2025.

What is an OCI card and who is eligible?

The OCI scheme allows eligible foreign nationals of Indian origin to register as Overseas Citizens of India. But OCI does not confer Indian citizenship.

Under Section 7A of the Citizenship Act, 1955, eligibility extends to people who were Indian citizens when the Constitution commenced on January 26, 1950, or became citizens thereafter; those eligible for Indian citizenship at that time; people belonging to territories that became part of India after August 15, 1947; their children, grandchildren, and great-grandchildren. Certain minor children and foreign spouses of Indian citizens are also eligible, subject to statutory conditions. On the other hand, people who are or whose parents, grandparents, or great-grandparents are or were citizens of Pakistan, Bangladesh, or any other country notified by the Central government are ineligible.

Although the Immigration and Foreigners Act, 2025, does not expressly prohibit OCI cardholders from attending political rallies, it allows the government to restrict their activities and continued presence.



Actor Imran Khan being taken away by the police during the protest against the Chief Election Commissioner in Mumbai. EMMANUAL YOGINI

What privileges does an OCI card provide?

OCI cardholders receive a lifelong, multiple-entry visa to India and are generally exempt from registration with the FRRO, regardless of the duration of their stay.

They have parity with Indian nationals for domestic airfares and entry fees to national parks, wildlife sanctuaries, national monuments, historical sites, and museums. They also have parity with non-resident Indians (NRIs) in certain matters, including the purchase and sale of immovable property other than agricultural land, farmhouses, and plantations.

OCI cardholders can pursue certain professions such as medicine, dentistry, nursing, pharmacy, law, architecture, and chartered accountancy subject to certain laws. They may also appear for certain national entrance exams, including NEET and the JEE, but only for NRI or supernumerary seats where applicable.

However, they require special permission for missionary work, tabligh activities, mountaineering, and journalism; and also for certain internships, employment involving foreign diplomatic missions or governments, and visits to protected or restricted areas.

Do OCI cardholders have political rights?

No. Under Section 7B of the Citizenship Act, OCI cardholders cannot vote or contest elections. They are also ineligible for several constitutional offices, including the President, Vice-President, and judges of the Supreme Court and High Courts, as well as most public employment reserved for Indian citizens. Article 19 of the Constitution, which guarantees freedoms including speech and peaceful assembly, applies specifically to citizens. OCI cardholders cannot claim these fundamental rights.

Can they participate in political protests?

The Immigration and Foreigners Act, 2025, empowers the government to regulate foreign nationals' presence and activities in India. Although it does not expressly prohibit them from attending political rallies, it allows the government to restrict their activities and continued presence.

The notice issued to Mr. Khan invokes Section 7(2)(c), read with Section 29. Section 7 empowers the government to regulate or restrict the entry, departure, and continued presence of foreigners. Under Section 7(2)(c), the government may direct a foreigner not to remain in India or in a specified area. While Section 7 also allows the government to restrict specified activities, the notice invokes its power to direct Mr. Khan to leave India.

Section 29 empowers the government to order the removal of a foreigner for violating the Act, its rules or orders issued under it, or on the basis of an adverse security report. It also authorises officers to exercise reasonable powers to enforce such directions.

In 2020, the Calcutta High Court quashed a notice directing Polish student Kamil Siedczynski to leave India after he participated in an anti-Citizenship (Amendment) Act rally in Kolkata. The court held that participation in a political rally by itself was insufficient to justify the action. It observed that although Article 19 rights are specifically guaranteed to citizens, the right to have political views and participate in political activities falls under Article 21, which guarantees the right to life and personal liberty, which is available to foreigners as well.

However, the Calcutta High Court's judgment was delivered under the Foreigners Act, 1946, which has since been replaced by the Immigration and Foreigners Act, 2025.

Can the government cancel OCI registration?

Yes. Section 7D of the Citizenship Act empowers the Central government to cancel OCI registration on specified grounds. These include obtaining registration through fraud, false representation, or concealment of material facts, showing disaffection towards the Constitution, certain dealings with an enemy during a war involving India, and being sentenced to imprisonment for at least two years within five years of registration.

Registration may also be cancelled if the government considers it necessary in the interests of India's sovereignty and integrity, security, friendly relations with foreign countries, or the general public.

What are the remedies available?

Mr. Khan can make a representation to the FRRO or the Central government seeking reconsideration or withdrawal of the notice.

Section 7 of the Immigration and Foreigners Act does not set out a comparable appeal procedure against a direction to leave India. He can also approach the High Court under Article 226 of the Constitution, challenging the notice and seeking interim protection against its enforcement. Mr. Khan's lawyers have already approached the High Court.

Unless the FRRO withdraws or modifies the notice or a court grants interim protection, the direction remains operative.

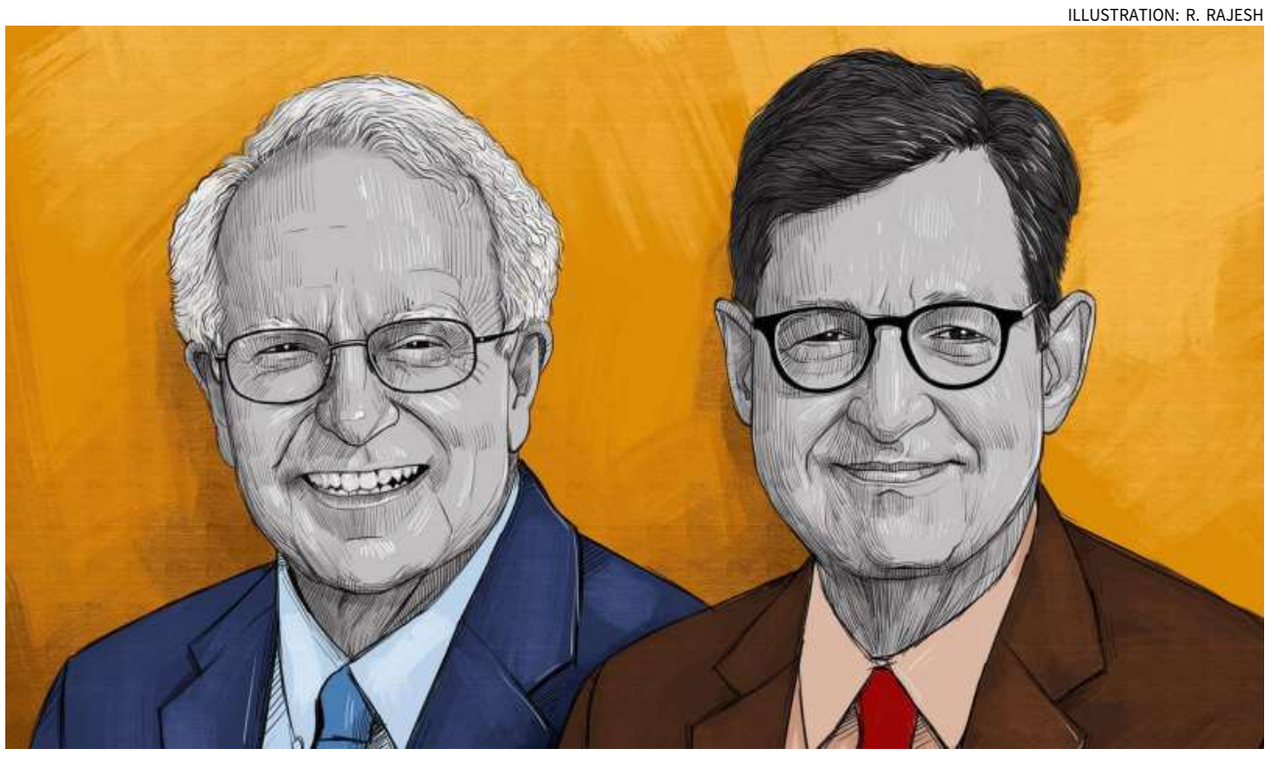
Putting people in economics

Robert E. Hall and Michael Woodford

The winners of the 2026 Economics Nobel Prize transformed macroeconomics by showing how individual choices, household expectations and business behaviour shape the economy and influence the effectiveness of government policies

T.C.A. Sharad Raghavan

ILLUSTRATION: R. RAJESH



Most people view macroeconomics as a result of the mechanical tweaking of some variables to alter other variables. For example, if aggregate consumer demand is down, there is immediately a call for lower interest rates and higher government doles. Similarly, when demand outstrips supply, then the call is the opposite: raise interest rates and cut back on handouts.

There was also a long-held belief that unemployment and inflation had an inverse relationship – as shown in the famous Phillips Curve – and that income and consumption moved in tandem. If governments could tweak one, the belief was that this would result in changes in the other.

It was this that Robert E. Hall and Michael Woodford turned on its head, for which they were awarded the 2026 Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel, commonly called the Economics Nobel. The economic thinking that the 'invisible hand' of the government should step in to stabilise the economy was born out of the landmark work of that titan of economic theory, John Maynard Keynes, and was put to great use in the aftermath of the Great Depression in the U.S. in the 1930s.

This stabilisation policy remained popular for several decades until it was upended by the oil shock the U.S. economy faced in the 1970s. During this crisis, the relationship between several macroeconomic variables broke down. That is, inflation rose at an accelerated pace even though unemployment was high, and the link between income and consumption looked like it was no longer holding.

What this meant was that conventional stabilisation policy no longer worked – government stimulus came at the wrong time, which had long-term detrimental effects on growth and inflation.

Economists such as Robert E. Lucas, the 1995 Economics Nobel Prize winner, then got to work to show that the dry mechanisms of simply working with macroeconomic variables were not the way. Inflation at the macro level was the result of pricing decisions by thousands of individual companies, and total private consumption was the result of actions taken by lakhs of individual households.

However, while it was increasingly accepted that households make consumption choices not only on the basis of their current situation but also on the basis of their future expectations, the difficulty arose in trying to quantify the effect of these future expectations.

Policy and behaviour

Enter Robert E. Hall. Born in 1943 in Palo Alto, California, Mr. Hall graduated from UC Berkeley in 1964 and earned his PhD from MIT. Even as a graduate student, he was already working on the issue of how government policy affects business behavior, research that eventually led to a seminal 1967 paper 'Tax Policy and Investment Behavior'.

He went on to serve on the economics faculty of UC Berkeley from 1967 to 1970 before moving to MIT and then to Stanford University, where he still works as the Robert and Carole McNeil Joint Hoover Senior Fellow and Professor of Economics.

Mr. Hall's greatest contribution, and for which he was awarded the Nobel Prize, however, had to do with the households not companies.

The central idea of his research, later put down in a landmark paper in 1978, was that if the consumption decisions of households is governed by their expectations of future income and other such factors, then their current consumption is also an accurate indicator of their future expectations. That is, if a household is optimistic about the future, it will consume more than if it were pessimistic.

A related observation was that households will also only ever change their consumption patterns if their assessment of the future changes. In turn, their assessment of the future changes when they obtain new and relevant information. Further, since new information cannot be predicted by definition, changes in consumption also cannot be predicted.

Now, here is the key for policymakers. If the rule is that changes in consumption cannot be predicted because they are based on unpredictable new information, then situations where it can be predicted must be because households are breaking this rule. This breaking of the rules usually takes place when a household is living paycheck to paycheck, or hand-to-mouth. They cannot borrow and have no savings, and so they spend only exactly what they earn. That is, their consumption is entirely predictable.

Using this finding, and the empirical work done by Mr. Hall, policymakers could now measure in exact terms the percentage of the population that was living hand-to-mouth.

The other major contribution of Mr. Hall's work was bringing in a 'human' element to macroeconomics, providing policymakers a map of sorts to how individuals react to both fear and confidence about the future.

Mr. Hall's work addressed the demand side, giving macroeconomics microeconomic underpinnings. The supply side theory was a result of the work done by Michael Woodford.

Mr. Woodford was born in 1955 in Massachusetts and he went on to complete his undergraduate studies at the University of Chicago, earn a Juris Doctor from Yale Law School, and complete his PhD in Economics from MIT. Not only did Mr. Woodford build on Mr. Hall's demand side contributions, but he also brought about a decisive change in how inflation was viewed. That is, in a series of papers starting in the 1990s, he showed that inflation is based on the behaviour of individual companies that set prices, knowing that it would be too difficult to change these prices every time demand or input costs change. Firms are not only future-looking in their behaviour, but their prices also tend to be 'sticky' – they don't change quickly.

Looking to the future

Now, the follow-up of this for policymakers was that, since businesses look to the future, their expectations about future interest rates are very important to manage inflation. As a result, it becomes of prime importance for central bank policies to be credible and transparent.

That is why the RBI Governor, while announcing interest rate decisions, puts them in the context of the global and Indian economies, and gives the rationale and outlook in detail. This policy of transparency is followed by all major central banks around the world and has its foundations in Mr. Woodford's work.

Other aspects of Mr. Woodford's work also provided policymakers more effective tools to deal with supply shocks than they had during the 1970s, such as using monetary policy to do the job rather than fiscal policy.

Mr. Woodford combined all of this with Mr. Hall's demand side theory and created a complete framework, which he published in a 2003 monograph titled *Making Things Happen: A Theory of Causal Explanation*. This monograph laid the groundwork for understanding business cycles, what causes them, how they play out, their impact on stabilisation policy, and the importance of policy transparency.

In essence, the two economists provided the theoretical and empirical underpinnings to a basic but powerful idea for policymakers: the economy is made out of the people in it.

THE GIST

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